

An Introduction to Statistical Learning with Applications in R

Chapter 3 Exercises

We will start by loading the required libraries:

```
library("tidyverse")
library("krulRutils")
library("here")
library("ISLR2")
library("cowplot")
library("rlang")
options(scipen = 999) # Disable scientific notation
```

Simple Linear Regression

For the first part of this report, we will create a plot showing the relation between the variables contained in the “Advertising” dataset. We start by loading and cleaning the dataset:

```
advertising_tbl <- read_csv(here("data", "Advertising.csv")) %>%
  select(-1)
```

We now create a function that generates a scatter plot for one of the explanatory variables against the response variable “sales”.

```
make_advertising_plot <- function(var) {
  var_name <- enquo(var) |>
    as_name()
  advertising_tbl %>%
    mutate(
```

```

    fitted = fitted(lm(reformulate(var_name, response = "sales"), data = .))
  ) %>%
  ggplot(aes(x = {{var}}, y = sales)) +
  geom_smooth(method = "lm", se = TRUE, color = "#1f77b4") +
  geom_segment(
    aes(xend = {{var}}, yend = fitted),
    color = "#1f77b4",
    alpha = 0.5
  ) +
  geom_point(color = "#d62728", size = 1, alpha = 1) +
  theme(
    panel.grid.major = element_line(color = "grey80"),
    panel.grid.minor = element_line(color = "gray80")
  ) +
  labs(
    title = paste(str_to_title(var_name), "Advertising vs Sales"),
    x = paste(str_to_title(var_name), "Advertising Budget"),
    y = "Sales"
  )
}

```

With this function in place, we can now generate the individual plots for each explanatory variable:

```

advertising_plot <- list(
  make_advertising_plot(TV),
  make_advertising_plot(radio),
  make_advertising_plot(newspaper)
)

```

We now put all of these plots together using “plot_grid” from the “cowplot” package:

```

main_advertising_plot <- plot_grid(plotlist = advertising_plot, ncol = 3)

```

```

`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'
`geom_smooth()` using formula = 'y ~ x'

```

Finally, we display these plots:

```

final_plot <- (ggdraw() +
  draw_label(
    "Comparative Analysis of Advertising Investement on
    Tv, Radio, or Newspaper",
    fontface = "bold",
    x = 0.5,
    y = 0.93,
    hjust = 0.5,
    size = 12
  ) +
  draw_plot(main_advertising_plot, y = 0, height = 0.82)) %>%
print()

```

**Comparative Analysis of Advertising Investement on
Tv, Radio, or Newspaper**

